PART-A

PROGRAM-1

Q:Draw E-R diagram for the given scenario.

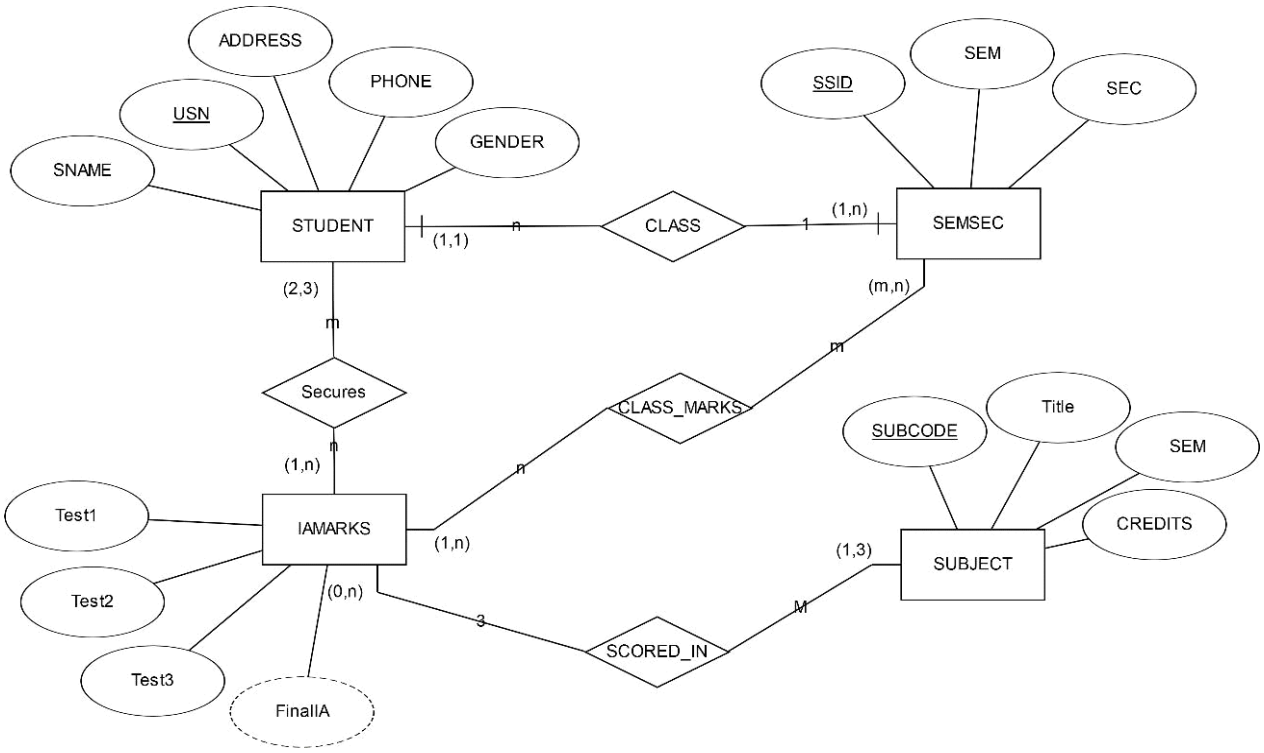
SOLUTION: Consider College and Company Database

1.COLLEGE DATABASE:

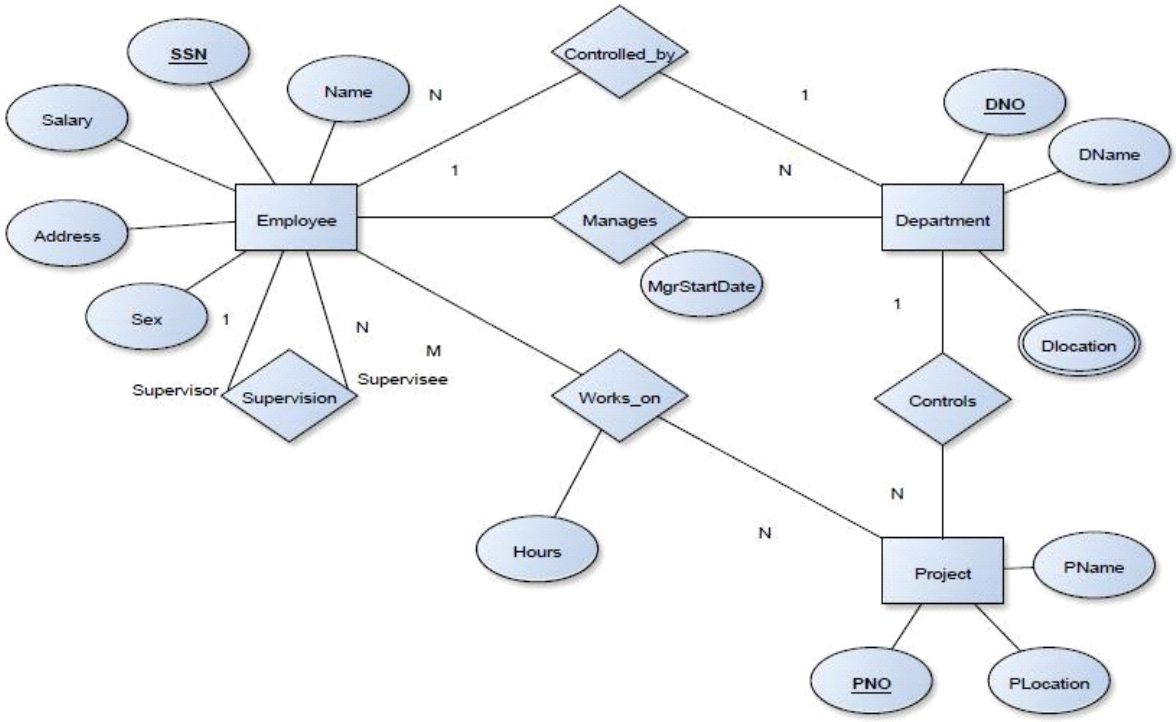
- STUDENT (USN,Sname,Address,Phone,Gender)
- SEMSEC(SSID,Sem,Sec)
- CLASS(USN,SSID)
- SUBJECT(Subcode,Title,Sem,Credits)
- IA MARKS(USN,Subcode,SSID,Test1,Test2,Test3,Final IA)

2.COMPANY DATABASE:

- EMPLOYEE(SSN,Name,Address,Sex,Salary,SuperSSN,Dno)
- DEPARTMENT(DNo,DNAME,MGRSSN,MgrStartDate)
- DLOCATION(DNo,DLoc)
- PROJECT(PNo,PName,PLocation,DNo)
- WORKS_ON(SSN,PNo,Hours)
    - **COLLEGE DATABASE: E-R DIAGRAM**



- **COMPANY DATABASE:-E R DIAGRAM**



PROGRAM-2

Q:Write queries for the following:

Creating user ,Creating a user , Viewing all Tables in a user, Creating Tables (With and Without Constraints), Inserting/Updating/Deleting Records in a Table, Saving (Commit) and Undoing (rollback)

Solution: Create following tables

Product(PID,Number,Name,Text,Price)

Purchase(PO,Number,PID,QTY,)

Steps :-

1. Create a user (user name: 3BSC, password: 3bsc)
2. Create table Product and Purchase with and without constraint
3. View all tables in inventory database
4. Insert five tuples into each relation
5. Display all the tuples in the product and purchase table
6. Update the product name for the PID=40 as CAMERA
7. Delete information about the product whose PID=50
8. Perform saving and undoing

Step1:- Create a user (user name: 3BSC, password: 3bsc)

CREATE USER 3BSC identified by 3bsc;

User Created.

GRANT ALL privileges to 3BSC;

Statement processed.

(log out and log in to 3BSC)

Step2:- Create table Product and Purchase with and without constriant

CREATE TABLE PRODUCT(PID NUMBER(10) PRIMARY KEY,NAME VARCHAR2(20) NOT NULL,PRICE NUMBER(8,2));

TABLE CREATED.

CREATE TABLE PURCHASE(PO NUMBER(10) PRIMARY KEY,PRODUCT_ID NUMBER(10) REFERENCES PRODUCT(PID) ,QTY NUMBER(5));

TABLE CREATED.

DESC PRODUCT;

NAME	NULL?	TYPE
PID	NOTNULL	NUMBER10
NAME	NOTNULL	VARCHAR2(20)
PRICE		NUMBER(8,2)

DESC PURCHASE;

NAME	NULL?	TYPE
PO	NOT NULL	NUMBER(10)
PRODUCT_ID		NUMBER(10)
QTY		NUMBER(5)

Step 3:-View all tables in inventory database

SELECT TABLE_NAME,STATUS FROM USER_TABLES;

TABLE_NAME	STATUS
PRODUCT	VALID
PURCHASE	VALID

Step 4:- Insert five tuples into each relation

Insert into Product(PID,NAME,PRICE)values(10,'PRINTER',20000);

1 ROW INSERTED

Insert into Product(PID,NAME,PRICE)values(20,'KEYBOARD',20000);

1 ROW INSERTED

Insert into Product(PID,NAME,PRICE)values(30,'MONITOR',15000);

1 ROW INSERTED

Insert into Product(PID,NAME,PRICE)values(40,'TABLE',25000);

1 ROW INSERTED

Insert into Product(PID,NAME,PRICE)values(50,'SCANNER',14000);

1 ROW INSERTED

Insert into Purchase(PO,PRODUCT_ID,QTY)values(101,10,25);

1 ROW INSERTED

Insert into Purchase(PO,PRODUCT_ID,QTY)values(102,40,20);

1 ROW INSERTED

Insert into Purchase(PO,PRODUCT_ID,QTY)values(107,30,40);

1 ROW INSERTED

Insert into Purchase(PO,PRODUCT_ID,QTY)values(104,40,50);

1 ROW INSERTED

Insert into Purchase(PO,PRODUCT_ID,QTY)values(105,40,10);

1 ROW INSERTED

Step 5:- Display all the tuples in the Product and Purchase table

SELECT * FROM PRODUCT;

PID	NAME	PRICE
10	PRINTER	20000
20	KEYBOARD	20000

30	**MONITOR**	**15000**
40	**TABLE**	**25000**
50	**SCANNER**	**14000**

SELECT * FROM PURCHASE;

PO	PRODUCT_ID	QTY
101	**10**	**25**
102	**40**	**20**
107	**30**	**40**
104	**40**	**50**
105	**40**	**10**

Step 6:- Update the Product name PID=40 as CAMERA

UPDATE PRODUCT SET NAME='CAMERA' WHERE PID=40;

1 ROW UPDATED.

SELECT * FROM PRODUCT;

PID	NAME	PRICE
10	**PRINTER**	**20000**
20	**KEYBOARD**	**20000**
30	**MONITOR**	**15000**
40	**CAMERA**	**25000**
50	**SCANNER**	**14000**

Step 7:- Delete information about the product whose PID=50

DELETE FROM PRODUCT WHERE PID=50;

1 ROW DELETED

SELECT * FROM PRODUCT;

PID	NAME	PRICE
10	PRINTER	20000
20	KEYBOARD	20000
30	MONITOR	15000
40	CAMERA	25000

Step 8:-Perform Saving and Undoing

COMMIT;

Statement processed.

Insert into PRODUCT(PID,NAME,PRICE)values(50,'mobile',35000);

1 row inserted

Insert into PRODUCT(PID,NAME,PRICE)values(60,'laptop',70000);

1 row inserted

SELECT * FROM PRODUCT;

PID	NAME	PRICE
10	PRINTER	20000
20	KEYBOARD	20000
30	MONITOR	15000
40	CAMERA	25000
50	MOBILE	35000
60	LAPTOP	70000

ROLLBACK;

Statement processed.

Select * from product;

PID	NAME	PRICE
10	PRINTER	20000
20	KEYBOARD	20000
30	MONITOR	15000
40	CAMERA	25000

PROGRAM 3 :- Alter a table,dropping/Truncating/Renaming a table,Backing up/restoring a database

Solution: Consider the library database with the following data and execute the queries

Lib(BID:Number,Title:Text,Publication:Text;Year_Of_Publication:Text)

Steps :-

1.create Lib table by properly specifying the constraint

2.rename the LIB as LIBRARY

3.Add a new column Price with Not Null constraint to the existing table library

4.All the constraints and the views that references the coloumn are dropped automatically along with the column.

5.Rename the BID to BOOKID in the library table

6.Change the data type of the column Year_Of _Publication as Text with size 15.

7.insert data into Library table .

8.Truncate table to delete the records

9.Drop table

Step 1:-create Lib table by properly specifying the constraint .

CREATE TABLE LIB(BID VARCHAR2(8) PRIMARY KEY,TITLE VARCHAR2(20) NOT NULL,AUTHOR VARCHAR2(20), PUBLICATION VARCHAR2(20), YEAR_OF_PUBLICATION NUMBER(4));

Table created.

DESC LIB;

Name	Null?	Type
BID	NOT NULL	VARCHAR2(8)
TITLE	NOT NULL	VARCHAR2(20)
AUTHOR		VARCHAR2(20)
PUBLICATION		VARCAHR2(20)
YEAR_OF_PUBLICATION		NUMBER(4)

Step2:-Rename the LIB as LIBRARY

ALTER TABLE LIB RENAME TO LIBRARY ;

Table altered.

DESC LIBRARY;

NAME	NULL?	TYPE
BID	NOT NULL	VARCHAR2(8)
TITLE	NOT NULL	VARCHAR2(2)
AUTHOR		VARCAHR2(20)
PUBLICATION		VARCHAR2(20)
YEAR_OF_PUBLICATION		NUMBER(4)

Step 3:- add a new column Price with Not Null constraint to the existing table library

ALTER TABLE LIBRARY ADD PRICE NUMBER(8,2) NOT NULL;

Table altered,

DESC LIBRARY;

NAME	NULL?	TYPE
BID	NOT NULL	VARCHAR2(8)
TITLE	NOT NULL	VARCHAR2(20)
AUTHOR		VARCHAR2(20)
PUBLICATION		VARCHAR2(20)
YEAR_OF_PUBLICATION		NUMBER(4)
PRICE	NOT NULL	NUMBER (8,2)

Step 4:- All the constraints and views that references the column are dropped automatically along with the column

ALTER TABLE LIBRARY DROP COLUMN AUTHOR CASCADE CONSTRAINTS;

Table altered.

DESC LIBRARY;

NAME	NULL?	TYPE
BID	NOT NULL	VARCAHR2(8)
TITLE	NOT NULL	VARCHAR2(20)
PUBLICATION		VARCHAR2(20)
YEAR_OF_PUBLICATION		NUMBER(4)
PRICE	NOT NULL	NUMBER(8,2)

Step 5:- rename the BID to BOOKID in the library table

ALTER TABLE LIBRARY RENAME COLUMN BID TO BOOKID;

Table altered.

DESC LIBRARY;

NAME	NULL?	TYPE
BOOKID	NOT NULL	VARCHAR2(8)
TITLE	NOT NULL	VARCHAR2(20)
PUBLICATION		VARCHAR2(20)
YEAR_OF_PUBLICATION		NUMBER(4)
PRICE	NOT NULL	NUMBER(8,2)

STEP 6:-Change the data type of the column YEAR_OF_PUBLICATION as Text with size 15

ALTER TABLE LIBRARY MODIFY YEAR_OF_PUBLICATION VARCHAR2(15);

Table altered.

DESC LIBRARY;

NAME	NULL?	TYPE
BOOKID	NOT NULL	VARCHAR2(8)
TITLE	NOT NULL	VARCHAR2(20)
PUBLICATION		VARCHAR2(20)
YEAR_OF_PUBLICATION		VARCHAR2(15)
PRICE	NOT NULL	NUMBER(8,2)

Step 7:- Insert data into Library table

INSERT INTO LIBRARY VALUES('SP001','DBMS','SKYWARD_PUBLISHERS','2022',300);

1 row inserted

SELECT * FROM LIBRARY;

BOOKID	TITLE	PUBLICATION	YEAR_OF_PUBLICATION	PRICE
SP001	DBMS	SKYWARD_PUBLISHERS	2022	300

STEP 8:- Truncate table to delete the records

TRUNCATE TABLE LIBRARY;

Table LIBRARY truncated.

FOR VERIFICATION

SELECT * FROM LIBRARY;

No data found

Step 9:- Drop table

DROP TABLE LIBRARY;

Table dropped

FOR VERIFICATION

DESC LIBRARY;

ORA-04043:-object LIBRARY does not exist

Program 4: for a given set of relation schemes, create tables and perform the following simple queries ,Simple Queries with Aggregate functions, Queries with Aggregate functions (group by and having clause)

Solution:

Consider the salary database and execute the following simple queries

SALARYDB(EID:String;NAME:Text ;DEPT:String,DOJ:Date;Salary:Number)

Steps:

1.create table SALARYDB

2.enter five tuples into the table

3.Display Employee Number and their Salary

4.Find the sum of salaries of all the employees

5.Find the sum and average salaries of employees of a particular department.

6.Find the number of the employees working for each department .

7.Display Employee information in ascending and descending order of their date of joining .

8.Find the highest salary that an employee draws

9. Find the least salary that an employee draws

10.Display the details of the employee whose name is Rushank and salary is greater than 50000.

Step1:- Create table salary

CREATE TABLE SALARYDB (ENO VARCHAR2(8) PRIMARY KEY,NAME VARCHAR2(15) NOT NULL, DEPT VARCHAR2(10),DOJ DATE ,SALARY NUMBER(10,2));

Table created.

DESC SALARYDB;

NAME	NULL?	TYPE
ENO	NOT NULL	VARCHAR2(8)
NAME	NOT NULL	VARCHAR2(15)
DEPT		VARCHAR2(10)
DOJ		DATE
SALARY		NUMBER(10,2)

Step2 : enter five tuples into the table

INSERT INTO SALARYDB VALUES('SC1010','AHANA','HR','15-FEB-2010',60000);

1 row inserted

INSERT INTO SALARYDB VALUES('SC1011','RAMESH','FINANCE','10-MAR-2012',45000);

1 row created

INSERT INTO SALARYDB VALUES('SC1013','NAVEEN','MARKETING','8-JAN-2009',55000);

1 row created

INSERT INTO SALARYDB VALUES('SC1014','ANAGA','HR','14-APR-2012',35000);

1 row created

INSERT INTO SALARYDB VALUES('SC1015','RUSHANK','ADMIN','16-MAY-2011',55000);

1 row inserted .

INSERT INTO SALARYDB VALUES('SC1016','RUSHANK','FINANCE','4-JUN-2008',25000);

1 row inserted.

SELECT *FROM SALARYDB;

ENO	NAME	DEPT	DOJ	SALARY
SC 1010	AHANA	HR	15-FEB-10	60000
SC1011	RAMESH	FINANCE	10-MAR-12	45000
SC1013	NAVEEN	MARKETING	08-JAN-09	55000
SC1014	ANAGA	HR	14-APR-10	35000
SC1015	RUSHANK	ADMIN	16-MAY-11	55000
SC1016	RUSHANK	FINANCE	04-JUN-08	25000

Step 3 Display Employee Number and their Salary

SELECT ENO,SALARY FROM SALARYDB;

ENO	SALARY

SC 1010	**60000**
SC1011	**45000**
SC1013	**55000**
SC1014	**35000**
SC1015	**55000**
SC1016	**25000**

6 row selected

Step 4: Find the sum of salaries of all the employees

SELECT SUM(SALARY) AS "TOTAL_SALARY" FROM SALARYDB;

TOTAL_SALARY
275000

Step5 :find the sum and average salaries of the employees of a particular department

SELECT DEPT,SUM(SALARY) AS "TOTAL_SALARY",AVG(SALARY)AS AVG_SALARY" FROM SALARYDB GROUP BY DEPT;

DEPT	**TOTAL SALARY**	**AVERAGE SALARY**
ADMIN	**55000**	**55000**
HR	**95000**	**47500**
FINANCE	**70000**	**35000**
MARKETING	**55000**	**55000**

Step 6: find the number of employees working for each department

SELECT DEPT,COUNT(*)AS "NUMBER_OF_EMPLOYEES" FROM SALARYDB GROUP BY DEPT;

DEPT	**NUMBER_OF_EMPLOYEES**
ADMIN	**1**

HR	**2**
FINANCE	**2**
MARKETING	**1**

Step 7: Display Employee information in ascending and descending order of their date of joining

SELECT * FROM SALARYDB ORDER BY DOJ ASC;

ENO	NAME	DEPT	DOJ	SALARY
SC1016	RUSHANK	FINANCE	04-JUN-08	25000
SC1013	NAVEEN	MARKETING	08-JAN-09	55000
SC1010	AHANA	HR	15-FEB-10	60000
SC1015	RUSHANK	ADMIN	16-MAY-11	55000
SC1011	RAMESH	FINANCE	10-MAR-12	45000
SC1014	ANAGA	HR	14-APR-12	35000

6 row selected

SELECT * FROM SALARYDB ORDER BY DOJ DESC;

ENO	NAME	DEPT	DOJ	SALARY
SC1014	ANAGA	HR	14-APR-12	35000
SC1011	RAMESH	FINANACE	10-MAR-12	45000
SC1015	RUSHANK	ADMIN	16-MAY-11	55000
SC 1010	AHANA	HR	15-FEB-10	60000
SC 1013	NAVEEN	MARKETING	08-JAN-09	55000
SC1016	RUSHANK	FINANCE	14-JUN-08	25000

6 row selected

Step 8: Find the highest salary that an employee draws .

SELECT MAX(SALARY)AS "HIGHEST_SALARY" FROM SALARYDB;

HIGHEST_SALARY
60000

STEP 9: Find the least salary that an employee draws

SELECT MIN(SALARY)AS "LEAST_SALARY" FROM SALARYDB;

LEAST_SALARY
25000

STEP 10: Display the details of the employee whose name is rushank and salary is greater than 50000

SELECT*FROM SALARYDB WHERE NAME='RUSHANK' AND SALARY>50000;

ENO	NAME	DEPT	DOJ	SALARY
SC1015	RUSHANK	ADMIN	16-MAY-11	55000

Program 8:- execute the following queries

For each department that has more than five employees, retrieve the department number and number of employees who are making salary more than 40000

SOLUTION:

(1) For each department that has more than five employees, retrieve the department number and number of employees who are making salary more than 40000

SELECT D.DNAME ,D.DNO,COUNT()AS "NO_OF_EMP" FROM EMP E,DEPT D WHERE E.DEPTNO =D.DNO AND E.SALARY>40000 AND D.DNO IN(SELECT DEPTNO FROM EMP GROUP BY DEPTNO HAVING COUNT ()>5) GROUP BY D.DNO ,D.DNAME;

No row selected

PROGRAM 9:

SQL> SELECT P.PNUMBER,P.PNAME,COUNT(*)AS"NO_OF_EMPLOYEES_WORKING"FROM PROJECT P,WORKS_ON W WHERE P.PNUMBER=W.PNO GROUP BY P.PNUMBER,P.PNAME HAVING COUNT(*)>2;

PNUMBER	PNAME	NO_OF_EMPLOYEES_WORKING
10	ERP	3
30	FINANCE	3

SQL> CREATE VIEW EMPLOYEE_DEPARTMENT AS(SELECT E.SSN,E.ENAME,E.SALARY,E.DNO,D.DNAME FROM EMPLOYEE E,DEPARTMENT D WHERE E.DNO=D.DNUMBER);

View created.

SQL> SELECT *FROM EMPLOYEE_DEPARTMENT;

SSN	ENAME	SALARY	DNO	DNAME
1001	ANIRUDH	45000	1	ACCOUNTS
1002	LAKSHMI	65000	2	ACCOUNTS
1003	MAHLAKSHMI	45000	3	RESEARCH
1004	RAMESH	52000	3	RESEARCH
1005	RAMYA	52000	4	RESEARCH
1006	ANU	78000	5	ADMIN
1007	ANUSHA	40000	1	ACCOUNTS
1008	ANUSHA	39000	1	ACCOUNTS

8 rows selected.

SQL> DROP VIEW EMPLOYEE_DEPARTMENT;

View dropped.

SQL> CREATE VIEW EMPLOYEE_VIEW AS (SELECT SSN,ENAME,SALARY FROM EMPLOYEE WHERE SALARY<=50000)WITH CHECK OPTION;

View created.

SQL> SELECT *FROM EMPLOYEE_VIEW;

SSN ENAME SALARY

SSN	ENAME	SALARY
1001	ANIRUDH	45000
1003	MAHALAKSHMI	50000
1007	ANUSHA	40000
1008	ANUSHA	39000

SQL> INSERT INTO EMPLOYEE_VIEW(SSN,ENAME,SALARY)VALUES(1011,'SNIGDH',39000);

1 row created.

SQL> SELECT*FROM EMPLOYEE_VIEW;

SSN	ENAME	SALARY
1001	ANIRUDH	45000
1003	MAHALAKSHMI	50000
1007	ANUSHA	40000
1008	ANUSHA	39000
1011	SNIGDH	39000

SQL>INSERT INTO EMPLOYEE_VIEW(SSN,ENAME,SALARY)VALUES(1012,’SMAYAN’,99999);

Consider the Company database with following Schema

EMPLOYEE (FNAME, MINIT, LNAME, SSN, BDATE,ADDRESS,SEX,SALARY, SUPERSSN, DNO) DEPARTMENT (DNAME, DNUMBER,MGRSSN,MSRSTARTDATE)

DEPT_LOCATIONS (DNUMBER, DLOCATION)

PROJECT (PNAME, PNUMBER, PLOCATION, DNUM)

WORKS_ON (ESSN, PNO,HOURS)

DEPENDENT (ESSN, DEPENDENT_NAME, SEX, BDATE, RELATIONSHIP)

1) TABLE CREATION;

SQL> CREATE TABLE DEPARTMENT(DNAME VARCHAR(20) NOT NULL,DNUMBER NUMBER(20) PRIMARY KEY,DLOCATION VARCHAR(20));

Table created.

SQL> DESC DEPARTMENT;

```
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNAME                                     NOT NULL VARCHAR2(20)
 DNUMBER                                   NOT NULL NUMBER(20)
 DLOCATION                                          VARCHAR2(20)
```

SQL> CREATE TABLE EMPLOYEE(ENAME VARCHAR(20)NOT NULL,SSN NUMBER(20) PRIMARY KEY , BDATE DATE,ADDRESS VARCHAR(20), SALARY

NUMBER(20) NOT NULL,DNO NUMBER (10) REFERENCES DEPARTMENT(DNUMBER),SEX VARCHAR(20));

Table created.

SQL> DESC EMPLOYEE;

```
 Name                                      Null?    Type
```

```
 ----------------------------------------- -------- ----------------------------
 ENAME                                     NOT NULL VARCHAR2(20)
 SSN                                       NOT NULL NUMBER(20)
 BDATE                                              DATE
 ADDRESS                                            VARCHAR2(20)
 SALARY                                    NOT NULL NUMBER(20)
 DNO                                                NUMBER(10)
 SEX                                                VARCHAR2(10)
SQL> CREATE TABLE DEPT_LOCATION(DNUMBER REFERENCES DEPARTMENT(DNUMBER) NOT NULL,
DLOCATION VARCHAR(20));

Table created.

SQL> DESC DEPT_LOCATION;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNUMBER                                   NOT NULL NUMBER(20)
 DLOCATION                                          VARCHAR2(20)

SQL> CREATE TABLE PROJECT(PNAME VARCHAR(20) NOT NULL,PNUMBER NUMBER(20) PRIMARY
KEY,PLOCATION VARCHAR(20),DNUM REFERENCES DEPARTMENT(DNUMBER));

Table created.

SQL> DESC PROJECT;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 PNAME                                     NOT NULL VARCHAR2(20)
 PNUMBER                                   NOT NULL NUMBER(20)
```

```
PLOCATION                                 VARCHAR2(20)
DNUM                                      NUMBER(20)
```

```
SQL> CREATE TABLE WORKS_ON(ESSN NUMBER(10) REFERENCES EMPLOYEE(SSN)  NOT NULL, PNO
NUMBER(20) REFERENCES PROJECT(PNUMBER) NOT NULL, HOURS NUMBER(3,1) NOT NULL);

Table created.

SQL> DESC WORKS_ON;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ESSN                                      NOT NULL NUMBER(10)
 PNO                                       NOT NULL NUMBER(20)
 HOURS                                     NOT NULL NUMBER(3,1)

SQL> INSERT INTO DEPARTMENT VALUES('ACCOUNTS',1,'JAYANAGAR');

1 row created.

SQL> INSERT INTO DEPARTMENT VALUES('ACCOUNTS',2,'VIJAYNAGAR');

1 row created.

SQL> INSERT INTO DEPARTMENT VALUES('RESEARCH',3,'VIJAYNAGAR');

1 row created.
```

```
SQL> INSERT INTO DEPARTMENT VALUES('RESEARCH',4,'NAGARBHAVI');

1 row created.

SQL> INSERT INTO DEPARTMENT VALUES('ADMIN',5,'NAGARBHAVI');

1 row created.

SQL> SELECT *FROM DEPARTMENT;

DNAME                DNUMBER DLOCATION
-------------------- ---------- --------------------
ACCOUNTS                    1 JAYANAGAR
ACCOUNTS                    2 VIJAYNAGAR
RESEARCH                    3 VIJAYNAGAR
RESEARCH                    4 NAGARBHAVI
ADMIN                       5 NAGARBHAVI


SQL> INSERT INTO EMPLOYEE VALUES('ANIRUDH',1001,'14-JAN-90','BANGALORE',45000,1,'M');

1 row created.

SQL> INSERT INTO EMPLOYEE VALUES('LAKSHMI',1002,'04-MAR-98','MYSORE',65000,2,'F');

1 row created.
```

```
SQL> INSERT INTO EMPLOYEE VALUES('MAHLAKSHMI',1003,'22-DEC-90','MANGALORE',50000,3,'F');

1 row created.

SQL> INSERT INTO EMPLOYEE VALUES('RAKESH',1004,'22-JUN-95','HUBLI',52000,3,'M');

1 row created.

SQL> INSERT INTO EMPLOYEE VALUES('RAMYA',1005,'22-JUN-98','DAVANGERE',52000,4,'F');

1 row created.

SQL> INSERT INTO EMPLOYEE VALUES('ANU',1006,'17-APR-91','SHIMOGA',78000,5,'F');

1 row created.

SQL> SELECT *FROM EMPLOYEE;

ENAME                SSN BDATE     ADDRESS              SALARY          DNO                          SEX
ANIRUDH               1001 14-JAN-90 BANGALORE             45000          1                            M

LAKSHMI              1002 04-MAR-98 MYSORE                65000
       2 F

MAHLAKSHMI             1003 22-DEC-90 MANGALORE             50000
       3 F


ENAME                SSN BDATE     ADDRESS              SALARY
```

```
-------------------- ---------- --------- -------------------- ----------
      DNO SEX
---------- ----------
RAKESH                  1004 22-JUN-95 HUBLI                  52000
        3 M

RAMYA                   1005 22-JUN-98 DAVANGERE              52000
        4 F

ANU                     1006 17-APR-91 SHIMOGA                78000
        5 F

SQL> INSERT INTO DEPT_LOCATION VALUES(1,'BANGALORE');

1 row created.

SQL> INSERT INTO DEPT_LOCATION VALUES(2,'BANGALORE');

1 row created.

SQL> INSERT INTO DEPT_LOCATION VALUES(3,'MANGALORE');

1 row created.

SQL> SELECT *FROM DEPT_LOCATION;

   DNUMBER      DLOCATION
---------- --------------------
        1 BANGALORE
```

```
        2 BANGALORE
        3 MANGALORE


SQL> INSERT INTO PROJECT VALUES('ERP','10','BANGALORE',1);

1 row created.

SQL> INSERT INTO PROJECT VALUES('BANKING','20','MYSORE',2);

1 row created.

SQL> INSERT INTO PROJECT VALUES('FINANCE','30','HASSAN',2);

1 row created.

SQL> INSERT INTO PROJECT VALUES('MARKET','40','RAICHUR',3);

1 row created.

SQL> INSERT INTO PROJECT VALUES('ANALYTICA','50','SHIMOGA',4);

1 row created.

SQL> DESC PROJECT;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 PNAME                                     NOT NULL VARCHAR2(20)
 PNUMBER                                   NOT NULL NUMBER(20)
```

```
 PLOCATION                                 VARCHAR2(20)
 DNUM                                     NUMBER(20)


SQL> INSERT INTO WORKS_ON VALUES(1001,10,4.5);

1 row created.

SQL> INSERT INTO WORKS_ON VALUES(1002,10,5);

1 row created.

SQL> INSERT INTO WORKS_ON VALUES(1003,30,3);

1 row created.

SQL> INSERT INTO WORKS_ON VALUES(1004,40,3.5);

1 row created.

SQL> SELECT *FROM WORKS_ON;

     ESSN       PNO      HOURS
---------- ---------- ----------
     1001        10       4.5
     1002        10         5
     1003        30         3
     1004        40       3.5
```

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<u>PROGRAM 5:-</u> Execute the following queries

(a)Show the resulting salaries if every employee working on the 'RESEARCH' departments is given a 10% raise

(b)Find the sum of the salaries of all employees of the 'ACCOUNTS' department as well as the maximum salary ,the minimum salary, and the average salary in this department

<u>SOLUTION:</u>

(a) Show the resulting salaries if every employee working on the 'RESEARCH' departments is given a 10% raise

SOLUTION:

```
SQL> SELECT E.SSN,E.ENAME,D.DNAME,1.1*E.SALARY AS"INC_SALARY" FROM EMPLOYEE
E,DEPARTMENT D WHERE E.DNO=D.DNUMBER AND D.DNAME='RESEARCH';

     SSN ENAME            DNAME              INC_SALARY
---------- -------------------- -------------------- ----------
    1003 MAHLAKSHMI         RESEARCH              55000
    1004 RAKESH             RESEARCH              57200
    1005 RAMYA              RESEARCH              57200
```

(b)Find the sum of the salaries of all employees of the 'ACCOUNTS' department as well as the maximum salary ,the minimum salary, and the average salary in this department

SOLUTION:

SQL> SELECT MAX(E.SALARY),MIN(E.SALARY),SUM(E.SALARY),AVG(E.SALARY) FROM EMPLOYEE E, DEPARTMENT D WHERE E.DNO=D.DNUMBER AND D.DNAME='ACCOUNTS';

MAX(E.SALARY) MIN(E.SALARY) SUM(E.SALARY) AVG(E.SALARY)

---------- ---------- ---------- ----------

65000 45000 110000 55000

Program 6:-execute the following queries

- **Retrieve the name of each employee controlled by department number 5(use EXISTS operator)**
- **Retrieve the name of each dept And number of employees working in each department which has at least 2 employees**

A) Retrieve the name of each employee controlled by department number 5(use EXISTS operator)

SOLUTION:

```
SQL> SELECT E.ENAME FROM EMPLOYEE E WHERE EXISTS(SELECT
D.DNUMBER FROM DEPARTMENT  D WHERE E.DNO=D.DNUMBER AND
E.DNO=5);

ENAME
--------------------
ANU
```

B) Retrieve the name of each dept And number of employees working in each department which has at least 2 employees

SOLUTION:

```
SQL> SELECT D.DNAME ,COUNT(*)FROM EMPLOYEE E,DEPARTMENT D  WHERE
E.DNO=D.DNUMBER  GROUP BY D.DNAME HAVING COUNT (*)>=2;

DNAME              COUNT(*)
-------------------- ----------
ACCOUNTS                    2
RESEARCH                    3
```

Program 7 :- execute the following queries

- **For each project retrieve the project number ,the project name ,and the number of employees who work on that project (use GROUP BY)**
- **Retrieve the name of the employee who born in the year 1990's**

<u>SOLUTION:</u>

(a) For each project retrieve the project number ,the project name ,and the number of employees who work on that project (use GROUP BY)

SOLUTION:

```
SQL>  SELECT P.PNUMBER ,P.PNAME ,COUNT(*)AS "NO _OF_EMPLOYEE" FROM
PROJECT P , WORKS_ON W WHERE P.PNUMBER=W.PNO GROUP BY
P.PNUMBER,P.PNAME ;

  PNUMBER PNAME              NO _OF_EMPLOYEE
---------- -------------------- ---------------
        10 ERP                           2
        30 FINANCE                       1
        40 MARKET                        1
```

(b) Retrieve the name of the employee who born in the year 1990's

```
SQL> SELECT ENAME,BDATE FROM EMPLOYEE WHERE BDATE LIKE '%-%-90';

ENAME              BDATE
-------------------- ---------
ANIRUDH            14-JAN-90
MAHLAKSHMI         22-DEC-90
```

Create the following tables with properly specifying Primary keys, Foreign keys and solve the following queries.

BRANCH (Branchid, Branchname, HOD)

STUDENT (USN, Name, Address, Branchid, sem)

BOOK (Bookid, Bookname, Authorid, Publisher, Branchid)

AUTHOR (Authorid, Authorname, Country, age)

BORROW (USN, Bookid, Borrowed_Date)

1. Perform the following: a. Viewing all databases, Creating a Database, Viewing all Tables in a Database, Creating Tables (With and Without Constraints), Inserting/Updating/Deleting Records in a Table, Saving (Commit) and Undoing (rollback) Execute the following Queries:

<u>Step1:-</u> Creating a Database

CREATE USER ARPITHA identified by ARPITHA.

2. User Created.
3. **GRANT ALL privileges to ARPITHA.**

PROGRAM 1: TABLE CREATION

SQL> CREATE TABLE BRANCH(BRANCHID NUMBER(5) PRIMARY KEY,BRANHNAME VARCHAR(10) NOT NULL, HOD VARCHAR(10));

Table created.

SQL> DESC BRANCH;

```
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 BRANCHID                                  NOT NULL NUMBER(5)
 BRANHNAME                                 NOT NULL VARCHAR2(10)
 HOD                                                VARCHAR2(10)
```

SQL> CREATE TABLE STUDENT(USN VARCHAR(15) PRIMARY KEY, NAME VARCHAR(15) NOT NULL, ADDRESS VARCHAR(30),BRANCHID NUMBER(10) REFERENCES BRANCH(BRANCHID), SEM VARCHAR(10));

Table created.

QL> DESC STUDENT;

```
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 USN                                       NOT NULL VARCHAR2(15)
 NAME                                      NOT NULL VARCHAR2(15)
 ADDRESS                                            VARCHAR2(30)
 BRANCHID                                           NUMBER(10)
 SEM                                                VARCHAR2(10)
```

SQL> CREATE TABLE AUTHOR(AUTHORID VARCHAR(15) PRIMARY KEY,AUTHORNAME VARCHAR(20) NOT NULL, COUNTRY VARCHAR(10), AGE NUMBER(5));

Table created.

SQL> DESC AUTHOR;

```
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 AUTHORID                                  NOT NULL VARCHAR2(15)
 AUTHORNAME                                NOT NULL VARCHAR2(20)
 COUNTRY                                            VARCHAR2(10)
 AGE                                                NUMBER(5)
```

SQL> CREATE TABLE BOOK(BOOKID VARCHAR(10) PRIMARY KEY, BOOKNAME VARCHAR(30) NOT NULL, AUTHORID VARCHAR(15) REFERENCES AUTHOR, PUBLISHER VARCHAR(10) NOT NUL

L, BRANCHID NUMBER(5) REFERENCES BRANCH);

Table created

SQL> DESC BOOK;

```
 Name                                      Null?    Type
```

```
 ----------------------------------------- -------- ----------------------------
 BOOKID                                    NOT NULL VARCHAR2(10)
 BOOKNAME                                  NOT NULL VARCHAR2(30)
 AUTHORID                                           VARCHAR2(15)
 PUBLISHER                                 NOT NULL VARCHAR2(10)
 BRANCHID                                           NUMBER(5)
SQL> CREATE TABLE BORROW(USN VARCHAR(19) REFERENCES STUDENT,BOOKID VARCHAR(10)
REFERENCES BOOK2(BOOKID),BORROW_DATE DATE,PRIMARY KEY(USN,BOOKID));


Table created.


SQL> DESC BORROW;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 USN                                       NOT NULL VARCHAR2(19)
 BOOKID                                    NOT NULL VARCHAR2(10)
 BORROW_DATE                                        DATE
```

<u>Program 1</u> : Viewing all databases, Creating a Databases, Viewing all Tables in a Database, Creating Tables (With and Without Constraints), Inserting/Updating/Deleting Records in a Table, Savings(Commit) and Undoing(rollback)

<u>Solution:-</u>

<u>Step 1:</u>

View all tables in COLLEGE Database

SELECT TABLE_NAME, STATUS FROM USER_TABLES;

TABLE_NAME	STATUS

BRANCH	VALID
STUDENT	VALID
AUTHOR	VALID
BOOK 2	VALID
BORROW	VALID

SQL> INSERT INTO BRANCH VALUES(10,’BCA’,’SANTHOSH S’);

1 row created.

SQL> INSERT INTO BRANCH VALUES(20,’BBA’,’RASHMI E’);

1 row created.

SQL> INSERT INTO BRANCH VALUES(30,’BCOM’,’RAMESH A’);

1 row created.

SQL> INSERT INTO BRANCH VALUES(40,’BSC’,’ADITYA P');

1 row created.

SQL> INSERT INTO BRANCH VALUES(50,’BA’,’ASHA P');

1 row created.

```
SQL> SELECT *FROM BRANCH;

  BRANCHID BRANHNAME  HOD

---------- ---------- ----------

        10 BCA        SANTHOSH S

        20 BBA        RASHMI E

        30 BCOM       RAMESH A

        40 BSC        ADITYA P

        50 BA         ASHA P
```

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202201','ANURADHA','JAYANAGAR',10,'II
SEM');
```

1 row created.

```
INSERT INTO STUDENT
VALUES('SCAS202202','MANJULA','BASAVANGUDI',10,'
II SEM');
```

1 row created.

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202203','MANJULA','BASAVANGUDI',10,'
II SEM');
```

1 row created.

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202203','MANJULA','BASAVANGUDI',10,'
II SEM');

SQL> INSERT INTO STUDENT
VALUES('SCAS202204','LAKSHMI','BASVANGUDI',10,'IV
SEM');
```

1 row created.

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202205','RENUKA','HANUMANTHANAGA
R',20,'II SEM');
```

1 row created.

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202206','ARUN','JPNAGAR',30,'II SEM');
```

1 row created.

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202207','ABHI','GIRINAGAR',40,'II SEM');
```

1 row created.

```
SQL> INSERT INTO STUDENT
VALUES('SCAS202208','DEEPTI','GIRINAGAR',50,'IV
SEM');
```

1 row created.

SQL> SELECT*FROM STUDENT;

USN NAME ADDRESS
BRANCHID

--- --- --- ---

SEM

SCAS202204 LAKSHMI BASVANGUDI
10

IV SEM

SCAS202205 RENUKA HANUMANTHANAGAR
20

II SEM

SCAS202206 ARUN JPNAGAR
30

II SEM

USN NAME ADDRESS
BRANCHID

--- --- --- ---

SEM

SCAS202207 ABHI GIRINAGAR
40

II SEM

SCAS202208 DEEPTI GIRINAGAR
50

IV SEM

SCAS202201 ANURADHA JAYANAGAR
10

II SEM

USN NAME ADDRESS
BRANCHID

--- --- --- ---

SEM

SCAS202202 MANJULA BASAVANGUDI
10

II SEM

SCAS202203 MANJULA BASAVANGUDI
10

II SEM

8 rows selected.

```
SQL> INSERT INTO AUTHOR
VALUES('NEPCOMP01','ARUNA','INDIA',36);
```

1 row created.

```
SQL> INSERT INTO AUTHOR
VALUES('NEPCOMP02','SUMA','INDIA',3);
```

1 row created.

```
SQL> INSERT INTO AUTHOR
VALUES('NEPCOMM02','SANGEETHA','INDIA',42);
```

1 row created.

```
SQL> INSERT INTO AUTHOR
VALUES('NEPCOMM01','DILIP','INDIA',39);

1 row created.

SQL> INSERT INTO AUTHOR
VALUES('NEPSC101','SHEKAR','INDIA',44);

1 row created.

SQL> SELECT*FROM AUTHOR;

AUTHORID       AUTHORNAME          COUNTRY
AGE

--------------- -------------------- ---------- ----------

NEPCOMP01      ARUNA               INDIA            36

NEPCOMP02      SUMA                INDIA            3

NEPCOMM02      SANGEETHA           INDIA            42
```

```
NEPCOMM01       DILIP              INDIA            39

NEPSC101        SHEKAR             INDIA            44
```

```
SQL> INSERT INTO BOOK
VALUES('NEPDBMS','DBMS','NEPCOMP02','SKYWARD',
10);

1 row created.

INSERT INTO BOOK
VALUES('NEPDBMS','DBMS','NEPCOMP02','SKYWARD',
10);

1 row created.

INSERT INTO BOOK
VALUES('NEPSE','SE','NEPCOMP02','SKYWARD',10);

1 row created.

SQL> INSERT INTO BOOK
VALUES('NEPJAVA','JAVA','NEPCOMM01','OXFORD',20);
```

1 row created.

```
INSERT INTO BOOK2
VALUES('NEPMATHS','MATHS','NEPSC101','OXFORD',30
);
```

1 row created.

```
SELECT * FROM BOOK;
```

```
BOOKID     BOOKNAME                    AUTHORID
PUBLISHER    BRANCHID

---------- ------------------------------ --------------- ---------- ----
------

NEPDBMS    DBMS                        NEPCOMP02
SKYWARD           10

NEPSE      SE                          NEPCOMP02
SKYWARD           10

NEPJAVA    JAVA                        NEPCOMM01
OXFORD            20

NEPMATHS   MATHS                       NEPSC101
OXFORD            30

NEPPHY     PHYSICS                     NEPCOMM02
SHREE             40
```

```
SQL> INSERT INTO BORROW
VALUES('SCAS202207','NEPMATHS','6-JUN-2022');
```

1 row created.

```
SQL> INSERT INTO BORROW
VALUES('SCAS202208','NEPMATHS','6-JUN-2022');
```

1 row created.

```
SQL> INSERT INTO BORROW
VALUES('SCAS202208','NEPDBMS','6-JUN-2022');
```

1 row created.

SQL> SELECT *FROM BORROW;

```
USN          BOOKID    BORROW_DA

--------------- ---------- ---------

SCAS202203     NEPDBMS    20-MAY-23

SCAS202205     NEPDBMS    20-MAY-23

SCAS202203     NEPMATHS   20-MAY-23

SCAS202203     NEPSE      20-MAY-22

SCAS202201     NEPDBMS    20-MAY-22

SCAS202207     NEPMATHS   06-JUN-22

SCAS202208     NEPMATHS   06-JUN-22

SCAS202208     NEPDBMS    06-JUN-22


8 rows selected.
```

Step 3 :

Updating Records in a Table.

UPDATE BOOK SET PUBLISHER='SKYWARD' WHERE BOOKID = 'NEPJAVA';

1 row updated.

QL> SELECT *FROM BOOK;

```
BOOKID     BOOKNAME                    AUTHORID       PUBLISHER    BRANCHID
---------- ------------------------------ --------------- ---------- ----------
NEPDBMS    DBMS                        NEPCOMP02      SKYWARD          10
NEPSE      SE                       NEPCOMP02      SKYWARD          10
NEPJAVA    JAVA                       NEPCOMM01      SKYWARD          20
NEPMATHS   MATHS                        NEPSC101       OXFORD          30
NEPPHY     PHYSICS                    NEPCOMM02      SHREE           40
```

Step 4 :

Deleting Records from a Table.

DELETE BORROW WHERE BOOKID='NEPPHY';

2 rows deleted.

```
SQL> SELECT *FROM BORROW;

USN            BOOKID     BORROW_DA
--------------- ---------- ---------
SCAS202203     NEPDBMS    20-MAY-23
SCAS202205     NEPDBMS    20-MAY-23
SCAS202203     NEPMATHS   20-MAY-23
SCAS202203     NEPSE      20-MAY-22
SCAS202201     NEPDBMS    20-MAY-22
SCAS202207     NEPMATHS   06-JUN-22
SCAS202208     NEPMATHS   06-JUN-22
SCAS202208     NEPDBMS    06-JUN-22

8 rows selected.
```

Step 5 :

Perform Saving (COMMIT)

```
SQL> INSERT INTO BRANCH VALUES(70,'MBA','VENKAT');

1 row created.

SQL> INSERT INTO BRANCH VALUES(60,'MCA','BALAJI');

1 row created.

SQL> SELECT *FROM BRANCH;
```

```
  BRANCHID BRANHNAME  HOD
---------- ---------- ----------
        10 BCA        SANTHOSH S
        20 BBA        RASHMI E
        30 BCOM       RAMESH A
        40 BSC        ADITYA P
        50 BA         ASHA P
        70 MBA        VENKAT
        60 MCA        BALAJI

7 rows selected.

SQL> COMMIT;

Commit complete.
```

Step 6 :

Perform Undoing (ROLL BACK)

```
SQL> INSERT INTO BRANCH VALUES(80,'MATHS','VEENA');

1 row created.

SQL> INSERT INTO BRANCH VALUES(90,'CHEMISTRY','JALAJA');

1 row created.

SQL> SELECT *FROM BRANCH;

  BRANCHID BRANHNAME  HOD
---------- ---------- ----------
        10 BCA        SANTHOSH S
        20 BBA        RASHMI E
        30 BCOM       RAMESH A
        40 BSC        ADITYA P
        50 BA         ASHA P
        70 MBA        VENKAT
        60 MCA        BALAJI
        80 MATHS      VEENA
        90 CHEMISTRY  JALAJA

9 rows selected.
```

```
SQL> ROLLBACK;

Rollback complete.

SQL> SELECT *FROM BRANCH;

  BRANCHID BRANHNAME  HOD
---------- ---------- ----------
        10 BCA       SANTHOSH S
        20 BBA       RASHMI E
        30 BCOM      RAMESH A
        40 BSC       ADITYA P
        50 BA        ASHA P
        70 MBA       VENKAT
        60 MCA       BALAJI

7 rows selected.
```

Program 2 :

(a) List the details of Students who are all studying in 2nd sem BCA.

(b) List the students who are not borrowed any books.

Solution:-

(a) List all details of Students who are all studying in 2nd sem BCA.

```
SQL> SELECT * FROM STUDENT S,BRANCH BR WHERE S.BRANCHID=BR.BRANCHID AND
S.SEM='II SEM' AND BR.BRANHNAME='BCA' ;

USN            NAME           ADDRESS                        BRANCHID
--------------- --------------- ------------------------------ ----------
SEM         BRANCHID BRANHNAME  HOD
---------- ---------- ---------- ----------
SCAS202201     ANURADHA       JAYANAGAR                        10
II SEM           10 BCA       SANTHOSH S

SCAS202202     MANJULA        BASAVANGUDI                       10
II SEM           10 BCA       SANTHOSH S

SCAS202203     MANJULA        BASAVANGUDI                       10
II SEM           10 BCA       SANTHOSH S
```

(b) List the students who are not borrowed any books

```
SQL> SELECT * FROM STUDENT S WHERE S.USN NOT IN(SELECT B.USN FROM BORROW B);

USN            NAME           ADDRESS                        BRANCHID
--------------- --------------- ------------------------------ ----------
SEM
----------
SCAS202204     LAKSHMI        BASVANGUDI                          10
IV SEM

SCAS202206     ARUN           JPNAGAR                             30
II SEM

SCAS202205     RENUKA         HANUMANTHANAGAR                     20
II SEM


USN            NAME           ADDRESS                        BRANCHID
--------------- --------------- ------------------------------ ----------
SEM
----------
SCAS202202     MANJULA        BASAVANGUDI                         10
II SEM
```

Program 3 :

(a)Display the USN, Student name, Branch_name, Book_name, Author_name, Books_Borrowed_Date of 2nd sem BCA Students who borrowed books.

(b)Display the number of books written by each Author.

Solution :-

(a) Display the USN, Student name, Branch_name, Book_name, Author_name, Books_Borrowed_Date of 2nd sem BCA Students who borrowed books.

```
SQL> SELECT S.USN,S.NAME,S.SEM,BR.BRANHNAME,
BK.BOOKNAME,A.AUTHORNAME,B.BORROW_DATE FROM STUDENT S,BRANCH BR,BOOK
BK,AUTHOR A,BORROW B WHERE S.BRANCHID=BR.BRANCHID AND S.BRANCHID=BK.BRANCHID
AND A.AUTHORID=BK.AUTHORID AND B.USN=S.USN AND BK.BOOKID=B.BOOKID AND S.SEM='II
SEM' AND BR.BRANHNAME='BCA';


USN           NAME          SEM       BRANHNAME

--------------- --------------- ---------- ----------

BOOKNAME                    AUTHORNAME         BORROW_DA

------------------------------ -------------------- ---------

SCAS202203     MANJULA        II SEM     BCA

SE                        SUMA              20-MAY-22


SCAS202201     ANURADHA       II SEM     BCA

DBMS                      SUMA              20-MAY-22
```

(b) Display the number of books written by each Author.

```
SQL> SELECT A.AUTHORNAME,COUNT(DISTINCT BK.BOOKID) AS "NO_OF_BOOKS" FROM
AUTHOR A,BOOK BK WHERE A.AUTHORID=BK.AUTHORID GROUP BY A.AUTHORNAME ;

AUTHORNAME          NO_OF_BOOKS
-------------------- -----------
SUMA                      2
DILIP                    1
SHEKAR                     1
SANGEETHA                    1
```

<u>Program 4 :</u>

(a)Display the Student details who borrowed more than two books.

(b)Display the student details who borrowed books of more than one Author.

Solution :-

(a) Display the student details who borrowed more than two books.

```
SQL> SELECT S.NAME FROM STUDENT S,BORROW B WHERE S.USN=B.USN GROUP BY S.NAME HAVING
COUNT(DISTINCT B.BOOKID)>2 ;

NAME
---------------
MANJULA
```

ANURADHA

(b) Display the student details who borrowed books of more than one Author.

```
SQL> SELECT S.NAME,COUNT(DISTINCT BK.AUTHORID) FROM STUDENT S,BOOK BK,BORROW B
WHERE S.USN=B.USN AND B.BOOKID=BK.BOOKID GROUP BY S.NAME HAVING
COUNT(DISTINCT BK.AUTHORID)> 1 ;

NAME          COUNT(DISTINCTBK.AUTHORID)
--------------- --------------------------
MANJULA                          2
DEEPTI                         2
```

Program 5 :

(a)Display the book names in descending order of their names.

(b)List the details of students who borrowed the books which are all published by the same publisher.

Solution :-

(a) Display the book names in descending order of their names.

```
SQL> SELECT * FROM BOOK ORDER BY BOOKNAME DESC;

BOOKID     BOOKNAME                    AUTHORID       PUBLISHER    BRANCHID
---------- ------------------------------ --------------- ---------- ----------
NEPSE      SE                       NEPCOMP02      SKYWARD          10
NEPPHY     PHYSICS                     NEPCOMM02      SHREE            40
NEPMATHS   MATHS                       NEPSC101       OXFORD           30
```

```
NEPJAVA    JAVA                    NEPCOMM01      SKYWARD          20
NEPDBMS    DBMS                      NEPCOMP02      SKYWARD          10
```

(b) List the details of students who borrowed the books which are all published by the same publisher.

```
SQL> SELECT S.NAME,COUNT(BK.PUBLISHER) FROM STUDENT S,BOOK BK,BORROW B WHERE
S.USN=B.USN AND B.BOOKID=BK.BOOKID GROUP
 BY S.NAME HAVING COUNT(BK.PUBLISHER)> 1;

NAME           COUNT(BK.PUBLISHER)
--------------- --------------------
MANJULA                         3
DEEPTI                        2
```

6) STUDENT (USN, name, date_of_birth, branch, mark1, mark2, mark3, total, GPA)

<u>Program 6 :</u> Perform the following:

Creating Tables (With and Without Constraints), Inserting / Updating / Deleting Records in a Table, Saving (Commit) and Undoing (rollback)

<u>Solution :-</u>

<u>Step 1:</u>

Creating Table (With and Without Constraints)

```
SQL> CREATE TABLE STUDENT1 (USN VARCHAR2(10) PRIMARY KEY,
NAME VARCHAR2(20) NOT NULL, DATE_OF_BIRTH DATE, BRANCH
VARCHAR2(10) NOT NULL, MARK1 NUMBER(4) NOT NULL, MARK2
NUMBER(4) NOT NULL, MARK3 NUMBER(4) NOT NULL, TOTAL
NUMBER(4), GPA NUMBER(4,2)) ;
w
Table created.

SQL> DESC STUDENT1;
```

```
Name                                      Null?    Type
----------------------------------------- -------- ----------------------------
USN                                       NOT NULL VARCHAR2(10)
NAME                                      NOT NULL VARCHAR2(20)
DATE_OF_BIRTH                                      DATE
BRANCH                                    NOT NULL VARCHAR2(10)
MARK1                                     NOT NULL NUMBER(4)
MARK2                                     NOT NULL NUMBER(4)
MARK3                                     NOT NULL NUMBER(4)
TOTAL                                              NUMBER(4)
GPA                                                NUMBER(4,2)
```

Step 2 :

Inserting Records into Table

```
SQL> INSERT INTO STUDENT1 VALUES('SCA202201','SANJANA','24-AUG-
2004','BCA',85,96,97,NULL,NULL);

1 row created.

SQL> INSERT INTO STUDENT1 VALUES('SCAC202201','ANIRUDH','10-OCT-
2004','BCOM',75,85,65,NULL,NULL);

1 row created.

SQL> INSERT INTO STUDENT1 VALUES('SCAB202201','AKASH','11-NOV-
2004','BBA',75,85,83,NULL,NULL);

1 row created.

SQL> INSERT INTO STUDENT1 VALUES('SCA202202','TANDRA',’1-DEC-
2004','BCA',84,56,63,NULL,NULL);

1 row created.
SQL> INSERT INTO STUDENT1 VALUES('SCA202203','ANUSHA',’1-JAN-
2004','BCA',68,72,78,NULL,NULL);

1 row created.
```

```
SQL> SELECT *FROM STUDENT1;

USN       NAME                 DATE_OF_B BRANCH        MARK1     MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3     TOTAL       GPA
---------- ---------- ----------
SCA202201  SANJANA             24-AUG-04 BCA              85        96
       97

SCAC202201 ANIRUDH             10-OCT-04 BCOM             75        85
       65

SCAB202201 AKASH               11-NOV-04 BBA              75        85
       83


USN       NAME                 DATE_OF_B BRANCH        MARK1     MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3     TOTAL       GPA
---------- ---------- ----------
SCA202202  TANDRA              01-DEC-04 BCA              84        56
       63

SCA202203  ANUSHA              01-JAN-04 BCA              68        72
       78
```

Step 3 :

Inserting Records into a Table and Saving(Commit)

```
SQL> INSERT INTO STUDENT1 VALUES('SCAC202203','SAHANA','15-JAN-
2005','BCA',45,65,68,NULL,NULL);

1 row created.

SQL> INSERT INTO STUDENT1 VALUES('SCAC202205','SAHANA','15-JAN-
2005','BCA',45,65,68,NULL,NULL);

1 row created.
SQL> SELECT *FROM STUDENT1;

USN       NAME                 DATE_OF_B BRANCH        MARK1     MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3     TOTAL       GPA
---------- ---------- ----------
```

```
SCA202201  SANJANA            24-AUG-04 BCA              85        96
      97

SCAC202201 ANIRUDH            10-OCT-04 BCOM             75        85
      65

SCAB202201 AKASH              11-NOV-04 BBA              75        85
      83


USN        NAME                DATE_OF_B BRANCH         MARK1      MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3      TOTAL       GPA
---------- ---------- ----------
SCA202202  TANDRA             01-DEC-04 BCA              84        56
      63

SCA202203  ANUSHA             01-JAN-04 BCA              68        72
      78

SCAC202203 SAHANA             15-JAN-05 BCA              45        65
      68


USN        NAME                DATE_OF_B BRANCH         MARK1      MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3      TOTAL       GPA
                                                  ---------- ---------- ----------
SCAC202205 SAHANA             15-JAN-05 BCA              45        65
      68


7 rows selected.
```

COMMIT;

Commit complete.

<u>Step 4 :</u>

Deleting Records into a Table and Saving (Commit)

SQL> DELETE STUDENT1 WHERE USN='SCAC202205';

1 row deleted.

Step 5 :

Undoing (rollback)

```
SQL>        INSERT
INTO    STUDENT1
VALUES('SCAB20
2204','SATYA','15-
MAR-
2004','BBA',68,77
,74,NULL,NULL);

1 row created.
```

```
SQL> INSERT INTO STUDENT1 VALUES('SCAB202205','DHANU','25-MAR-
2004','BBA',68,85,90,NULL,NULL);

1 row created.

SQL> ROLLBACK;

Rollback complete.

SQL> SELECT *FROM STUDENT1;

USN        NAME                 DATE_OF_B BRANCH         MARK1      MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3      TOTAL       GPA
---------- ---------- ----------
SCA202201  SANJANA              24-AUG-04 BCA               85        96
       97

SCAC202201 ANIRUDH              10-OCT-04 BCOM              75        85
       65

SCAB202201 AKASH                11-NOV-04 BBA               75        85
       83


USN        NAME                 DATE_OF_B BRANCH         MARK1      MARK2
---------- -------------------- --------- ---------- ---------- ----------
```

```
    MARK3      TOTAL       GPA
---------- ---------- ----------
SCA202202  TANDRA             01-DEC-04 BCA              84        56
       63

SCA202203  ANUSHA             01-JAN-04 BCA              68        72
       78

SCAC202203 SAHANA             15-JAN-05 BCA              45        65
       68


6 rows selected.
```

Step 6:

Updating Records in a Table.

```
SQL> UPDATE STUDENT1 SET
TOTAL=MARK1+MARK2+MARK3;

6 rows updated

SQL> SELECT *FROM STUDENT1;

USN       NAME                DATE_OF_B BRANCH         MARK1      MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3      TOTAL       GPA
---------- ---------- ----------
SCA202201  SANJANA            24-AUG-04 BCA              85        96
       97       278

SCAC202201 ANIRUDH            10-OCT-04 BCOM             75        85
       65       225

SCAB202201 AKASH              11-NOV-04 BBA              75        85
       83       243


USN       NAME                DATE_OF_B BRANCH         MARK1      MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3      TOTAL       GPA
---------- ---------- ----------
SCA202202  TANDRA             01-DEC-04 BCA              84        56
       63       203
```

```
SCA202203  ANUSHA            01-JAN-04 BCA             68        72
      78       218

SCAC202203 SAHANA            15-JAN-05 BCA             45        65
      68       178


6 rows selected.
```

Program 7 :

(a)Find the GPA score of all the students.

(b)Find the students who born on a particular year of birth from the date_of_birth column.

Solution:-

(a) Find the GPA score of all the students.

```
SQL> UPDATE STUDENT1 SET GPA=((100*TOTAL)/300)/10;

6 rows updated.


SQL> SELECT *FROM STUDENT1;

USN       NAME                 DATE_OF_B BRANCH        MARK1     MARK2
---------- -------------------- --------- ---------- ---------- ----------
    MARK3     TOTAL       GPA
---------- ---------- ----------
SCA202201  SANJANA           24-AUG-04 BCA             85        96
      97       278      9.27

SCAC202201 ANIRUDH           10-OCT-04 BCOM            75        85
      65       225       7.5

SCAB202201 AKASH             11-NOV-04 BBA             75        85
      83       243       8.1


USN       NAME                 DATE_OF_B BRANCH        MARK1     MARK2
```

```
---------- -------------------- --------- ---------- ---------- ----------
    MARK3      TOTAL       GPA
---------- ---------- ----------
SCA202202  TANDRA             01-DEC-04 BCA              84        56
       63       203      6.77

SCA202203  ANUSHA             01-JAN-04 BCA              68        72
       78       218      7.27

SCAC202203 SAHANA             15-JAN-05 BCA              45        65
       68       178      5.93


6 rows selected.
```

(b) Find the students who born on a particular year of birth from the date_of_birth column.

```
SQL> SELECT USN, NAME, BRANCH, DATE_OF_BIRTH FROM STUDENT1
WHERE DATE_OF_BIRTH LIKE '%-%-04';



USN       NAME               BRANCH     DATE_OF_B

---------- -------------------- ---------- ---------

SCA202201  SANJANA            BCA       24-AUG-04

SCAC202201 ANIRUDH            BCOM      10-OCT-04

SCAB202201 AKASH              BBA       11-NOV-04

SCA202202  TANDRA             BCA       01-DEC-04

SCA202203  ANUSHA             BCA       01-JAN-04
```

Program 8 :

(a)List the students who are studying in a particular branch of study.

(b)Find the maximum GPA score of the student branch- wise.

Solution :-

(a) List the students who are studying in a particular branch of study.

```
SQL> SELECT USN, NAME, BRANCH, DATE_OF_BIRTH FROM STUDENT1 WHERE BRANCH='BCA';

USN        NAME                 BRANCH     DATE_OF_B
---------- -------------------- ---------- ---------
SCA202201  SANJANA              BCA        24-AUG-04
SCA202202  TANDRA               BCA        01-DEC-04
SCA202203  ANUSHA               BCA        01-JAN-04
SCAC202203 SAHANA               BCA        15-JAN-05
```

(b) Find the maximum GPA score of the student branch-wise.

```
SQL> SELECT BRANCH, MAX(GPA) FROM STUDENT1 GROUP BY BRANCH;

BRANCH      MAX(GPA)
---------- ----------
BCA            9.27
BCOM             7.5
BBA             8.1
```